

UCLA Rocket Project

Vehicle: Prometheus (Hybrid) | Cycle: 25-26

Test Summary Report Cold Flow Campaign #1

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Test Campaign: Commercial Off-the-Shelf (COTS)
Test Location: Eng. IV Aerospace Design Laboratory
Test Date: 2026-02-03
Team: Propulsion Test Engineering
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Revision History

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1 Summary

On Tuesday, February 3rd, the subteam successfully cold-flowed the COTS engine to prepare for the hotfire campaign that weekend. Leading up to February 3rd, we successfully validated the fill table by pressurizing the table to 750 psi. However, after integrating the fill table with the rest of the COTS system and applying high pressure, the nylon lines blew. After purchasing new, higher-rated nylon lines and compatible PTCs, we successfully cold flowed on February 3rd around 10 PM.

2 Timeline

- Friday, January 30
1820 - 2100 - Low pressure test
2100 - 0000 - EGSE troubleshoot
- Saturday, January 31
Re-brief, repeated EGSE troubleshooting, propulsion departure, and EGSE departure.
- Sunday, February 1
New push-to-connect fittings. Explosions.
- Monday, February 2
Successful hydrostatic test, view report. New nylon lines, simplified hydrostatic article.
- Tuesday, February 3 - Coldflow

3 Issues Encountered

3.1 Push-to-Connect (PTC)

During early high-pressure testing, the push-to-connect fittings failed across multiple attempts. On the first attempt, PTCs sheared at the tank interface, and on the second attempt, a tube connection popped out entirely. The team sourced higher-rated PTCs from McMaster-Carr rated to 800 psi, which were brought in the following day. These failures caused repeated delays to the high-pressure schedule across the Saturday testing window.



(a) PTCs destroyed during high-pressure testing.



(b) Sheared PTC.

Figure 1: PTC failures during high-pressure testing.

3.2 Nylon Tubing Burst

During the first high-pressure attempt with the newly sourced PTCs installed, the nylon tubing in the system burst open. The tubing that failed was older stock found around the lab and in the COTS box, and its pressure rating had not been confirmed prior to use. A slight bend was also observed at the burst location, suggesting the geometry contributed to the failure. The team had 50 ft of 900 psi-rated nylon tubing on hand from McMaster-Carr as a backup, which was then swapped in as a replacement.

Figure 2: Burst nylon tubing.



3.3 PTC and Nylon Tubing Incompatibility

When the 900 psi-rated nylon tubing was installed into the higher-rated PTCs, the connections leaked significantly during low-pressure testing. Investigation revealed that the inner diameter of the new tubing was too tight for the PTC fittings, preventing a proper seal from forming. Worse,

when the team attempted to remove the tubing, it pulled out the internal plastic sealing components of the PTCs, rendering them all unusable. This left the team without compatible high-pressure fittings or tubing and required sourcing entirely new parts.

3.4 Pneumatic Line Burst on Third Attempt

During the third high-pressure attempt on Saturday night, the pneumatic lines burst again. The 900 psi-rated replacement lines that had been brought in were found to be incompatible with the PTCs available at that time. After significant team effort and an admin meeting to gather input from Ares members, a working combination of PTCs and tubing was identified. The new parts were validated via hydrostatic testing up to 900 psi (1.5x MEOP), clearing the path for a renewed high-pressure attempt and cold flow the following day.

4 What Went Well

4.1 Validated GSE Fill Table

GSE successfully high-pressed the fill table. Initially, there were frequent leaks, with a rate of around 5 psi/sec, but after tightening the hardline sleeves and the top stem of the ball valve, we passed the leak checks. We pressurized to 750 psi and were on standby to calculate the pressure drop (3.7 psi/min). From this, the prop leads decided to run the COTS system at 650 psi.

4.2 Verified Pressure Rating of Nylon Lines

We tried to high-pressure the entire COTS system (i.e., tank, fill table, engine, etc.) to mimic cold-flow conditions. However, during our attempt, the nylon lines burst, and we terminated the test. We were originally using old nylon lines with an unknown pressure rating, which was likely the cause. Afterward, we were ordered 50ft nylon lines rated for 900 psi.

4.3 Identifying Incompatibility of PTCs and Nylon Lines

After purchasing the new nylon lines, we connected them to the new PTCs. During leak checks, we found that the ID (inner diameter) of the nylon lines were too tight for the PTCs, resulting in an improper seal and frequent leaks. Additionally, removing the nylon lines broke the plastic sealing component from the PTCs. This meant we needed to purchase new PTCs that are compatible and try again.

4.4 Resolved Electronics

After the first attempt at high pressure, electronics resolved many issues, including successfully validating the wireless communication switch box and control box, confirming safe-state actuation of bimbas via the pneumatics box, and live DAQ for PTs.

5 Future Improvements

5.1 Purchase Correct PTCs and Nylon Lines

When purchasing PTCs and Nylon Lines, verify they are properly rated for high psi to prevent failed tests and to save money. Additionally, ensure the nylon tube IDs are compatible with our

PTCs.

5.2 Solenoid Overheating

The solenoids are rated for 12V. Use 12V, not a 24V power supply.

6 Prioritize Hydrostatic Pressure Tests

Water is incompressible, meaning that when a fitting or line fails under hydrostatic pressure, the stored energy releases almost instantaneously and locally. The result is a drip or a crack, not an explosion. Pressurized gas stores enormous elastic energy proportional to both pressure and volume. When a fitting fails under high-pressure gas, that energy releases violently and all at once, turning a manageable leak into a component failure.

The first weak link exposed was the nylon tubing. The lines used in the initial high-pressure attempt were old stock with an unconfirmed pressure rating. Had these lines been hydrostatically tested first, the burst would have occurred under water at low stored energy, flagging the lines as unfit. Instead, the failure occurred under high-pressure gas, resulting in a violent burst that immediately terminated the test and required a full line replacement.

The second weak link was the incompatibility between the 900 psi-rated nylon lines and the newly purchased PTCs. When the new lines were installed, every connection leaked due to the ID mismatch between the tubing and the fittings. Removing the lines then destroyed the internal plastic sealing components of the PTCs, rendering all of them unusable. A hydrostatic test on just the PTC-to-tubing connections would have identified the incompatibility immediately.

The third weak link was the pneumatic lines, which burst on the third high-pressure attempt after the replacement 900 psi lines were found to be incompatible with the PTCs on hand. Each of these failures shares the same root cause: the system was exposed to its most violent test condition before its components were individually validated under safe conditions.

Going forward, the standard procedure must be to hydrostatically test all articles in isolation before any high-pressure gas attempt. This means pressurizing to at least 1.5x MEOP and confirming zero leaks and no component deformation. The focus should be entirely on the interface connections, as the COTS tank is already pressure-rated. Skipping hydrostatic testing does not save time; it guarantees time will be lost later, under worse conditions, with more expensive consequences.